GCC Database Partner API

Amazon Developer Integration Guide

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1. Quick Start

Three steps to get your first data:

1 **Get your API key** from GCC (format: `gcc_partner_[hash]`). This full string, including the `gcc_partner_` prefix, is your `X-API-Key` value — use it exactly as provided in the header.

2 **Make your first request:**

```
# Test environment (current)
curl -X GET "https://gccwebapp.com/api/v1/certificates/gs1" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"

# Production environment (pending DNS configuration)
curl -X GET "https://greencirclecertified.com/database/api/v1/certificates/gs1" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

3 **You'll receive** a JSON response with all active certificates tagged to your partner account, including ASINs, certification details, and PDF download links.

> **Integration model:** You PULL data from GCC. GCC does not push to you. Call the API on your schedule to retrieve current certification data. Each API call returns the active certificate database. Amazon is responsible for identifying what is new or removed on their side to badge and unbadge products accordingly.

2. Authentication

Every request requires the `X-API-Key` header:

```
X-API-Key: gcc_partner_fc8498d5e8067766f74ed28bb439b17210196895e4e67828036f1737d222138a
```

Response Code	Meaning	Action
`200`	Success	Process the response
`401`	Invalid or missing API key	Check your key in the header
`403`	Endpoint not permitted	Contact GCC support
`404`	Resource not found	Verify certificate number or product ID
`429`	Rate limit exceeded	Wait and retry (200 API calls/hour)
`500`	Server error	Contact GCC support with request details

3. API Endpoints

Environment	Base URL	Status
Test (current)	`https://gccwebapp.com/api/v1`	Active — use this for development and integration testing
Production	`https://greencirclecertified.com/database/api/v1`	Pending DNS configuration

GET `/certificates/gs1` RECOMMENDED

List all certificates in GS1 JSON-LD format. ASINs and product IDs are always included.

Parameters: `pageNumber` (default: 1), `pageSize` (default/max: 100), `searchTerm`, `sortBy`

GET `/certificates/gs1/{certNumber}`

Get a single certificate by number in GS1 format.

Example: `/certificates/gs1/24-0366`

`GET` `/certificates`

List all certificates in standard JSON format.

Parameters: `pageNumber`, `pageSize`, `searchTerm`, `sortBy`, `includeProductIds` (boolean - set to `true` to include ASINs)

`GET` `/certificates/number/{certNumber}`

Get a single certificate by number in standard JSON format.

`GET` `/certificates/search/productid`

Find a certificate by ASIN, UPC, GTIN, or EAN.

Parameters: `productId` (required), `type` (ASIN, UPC, GTIN, EAN)

Example: `/certificates/search/productid?productId=B08XQRT456&type=ASIN`

Note: At a minimum, Amazon will be searching by ASIN.

`GET` `/certificates/{certNumber}/download`

Download the certificate PDF document. Returns `application/pdf`.

4. Response Data Model

Certificate Fields

Field	Type	Description	Example
`certificateNumber`	string	Unique certificate ID	`24-0366`
`status`	string	Always "Active" (only active certs returned)	`Active`
`issueDate`	datetime	Certificate effective date	`2024-01-15T00:00:00Z`
`expirationDate`	datetime	Certificate expiration date	`2025-01-15T00:00:00Z`
`company`	string	Certified company/brand name	`EcoProduct Co`
`productName`	string	Certified product name	`Recycled Paper Towels`
`certificationType`	string	Certification standard	`Sustainable Energy Practices`
`amazonCode`	string	Amazon's standardized cert identifier	`greencircle_sustainable_energy_practices`

Field	Type	Description	Example
`certificationResults`	string	Results achieved	`25%`
`pathway`	string	Certification pathway*	`Energy usage reduction`
`issuer`	string	Certifying body	`GreenCircle Certified`
`certificationCountries`	string	Marketplace countries**	`All`
`pdfDownloadUrl`	string	Relative URL to download cert PDF	`/api/v1/certificates/24-0366/download`

* Pathway will not be applicable for all certification types and will be left blank if not relevant.
** Marketplace IDs will not be passed. Amazon will apply the badge in all marketplaces where the product is sold, as the certificate is not restricted by location.

> **NOTE:** Only certificates that are currently active will be shared with Amazon. Any certificates that were previously badged but are no longer in the list/active should be unbadged. Amazon will manage the comparison and handle removals on their side.

Product Identifier Fields (within `productIds[]`)

Field	Type	Description	Example
`type`	string	ASIN, UPC, GTIN, EAN	`ASIN`
`value`	string	The product identifier value	`B08XQRT456`
`status`	string	Product ID status	`Active`
`partner`	string	Associated partner	`Amazon`
`productName`	string	Product variant name	`Paper Towels - 12 Pack`
`productSku`	string	Company SKU. **Note:** SKU is not always provided by the certificate holder. ASIN will always be passed at minimum. If SKU is not available, this field will be blank.	`RPT-12PK`

5. Request/Response Examples

5.1 List All Certificates (Standard JSON)

Request

```
GET /api/v1/certificates?includeProductIds=true
X-API-Key: YOUR_API_KEY
```

Response

```
{
  "success": true,
  "pagination": {
    "totalCount": 1,
    "pageNumber": 1,
    "pageSize": 100,
    "totalPages": 1,
    "hasNext": false,
    "hasPrevious": false
  },
  "data": [
    {
      "certificateNumber": "24-0366",
      "status": "Active",
      "issueDate": "2024-01-15T00:00:00Z",
      "expirationDate": "2025-01-15T00:00:00Z",
      "certificationCountries": "All",
      "partner": "Amazon",
      "certificationResults": "32%",
      "pathway": "Energy usage reduction",
      "company": "EcoProduct Co",
      "productName": "Energy-Efficient LED Bulb",
      "certificationType": "Sustainable Energy Practices",
      "amazonCode": "greencircle_sustainable_energy_practices",
      "issuer": "GreenCircle Certified",
      "productIds": [
        {
          "type": "ASIN",
          "value": "B08XQRT456",
          "status": "Active",
          "partner": "Amazon",
          "productName": "Energy-Efficient LED Bulb",
          "productSku": "LED-BULB-001"   // May be blank if not provided by certificate holder
        }
      ],
      "pdfDownloadUrl": "/api/v1/certificates/24-0366/download"
    }
  ]
}
```


5.2 List All Certificates (GS1 Format)

> **Note:** GS1 JSON-LD uses field names defined by the GS1 Web Vocabulary specification, which differ from the Standard JSON field names. For example, `productName` becomes `name` and `certificationType` becomes `program`. See Appendix A for the complete field mapping and an explanation of why the names differ.

Request

```
GET /api/v1/certificates/gs1
X-API-Key: YOUR_API_KEY
```

Response

Note: The response follows the GS1 Certification Data Model. Two fields (`amazonCode` and `pathway`) are GCC extensions not defined in the GS1 standard — they are included to support Amazon's Climate Pledge Friendly program mapping.

```
{
  "@context": "https://gs1us-technology.github.io/certification-data-model/cert_context.json",
  "$schema": "https://gs1us-technology.github.io/certification-data-model/cert_schema.json",
  "id": "https://greencirclecertified.com/database/api/v1/certificates/gs1/24-0366",
  "type": "CertificationReport",
  "issueDate": "2024-01-15",
  "issuer": "GreenCircle Certified",
  "products": [
    {
      "id": "https://id.gs1.org/01/00012345678901",
      "type": "Product",
      "gtin": "00012345678901",
      "name": "Energy-Efficient LED Bulb",                    // → productName
      "additionalTradeItemId": [                              // → productIds[]
        { "type": "ASIN", "value": "B08XQRT456" }
      ],
      "certificationInfo": {
        "type": "CertificationDetails",
        "agency": "GreenCircle Certified",                    // → issuer
        "agencyURL": "https://www.greencirclecertified.com",
        "program": "Sustainable Energy Practices",            // → certificationType
        "certificateId": "24-0366",                           // → certificateNumber
        "status": "ACTIVE",                                   // → status ("Active")
        "value": "32%",                                       // → certificationResults
        "startDate": "2024-01-15",                            // → issueDate
        "endDate": "2025-01-15",                              // → expirationDate
        "targetCountries": ["All"],                           // → certificationCountries
        "amazonCode": "greencircle_sustainable_energy_practices",  // → amazonCode (GCC extension)
        "pathway": "Energy usage reduction"                   // → pathway (GCC extension)
      }
    }
  ]
}
// See Appendix A for complete GS1 ⇔ Standard JSON field mapping
```

5.3 Search by ASIN

Request

```
GET /api/v1/certificates/search/productid?productId=B08XQRT789&type=ASIN
X-API-Key: YOUR_API_KEY
```

Response

```
{
  "success": true,
  "data": {
    "certificateNumber": "24-0512",
    "status": "Active",
    "certificationResults": "60%",
    "company": "CleanGreen Industries",
    "productName": "Recycled Paper Towels",
    "certificationType": "Recycled Content",
    "amazonCode": "greencircle_recycled_content",
    "productIds": [
      { "type": "ASIN", "value": "B08XQRT456", "productName": "Paper Towels - 6 Pack" },
      { "type": "ASIN", "value": "B08XQRT789", "productName": "Paper Towels - 12 Pack", "isMatchedId":
      { "type": "ASIN", "value": "B08XQRT999", "productName": "Paper Towels - 24 Pack" }
    ],
    "pdfDownloadUrl": "/api/v1/certificates/24-0512/download"
  }
}
```


> **Note:** Searching for one ASIN returns the full certificate with ALL associated product IDs. The `isMatchedId: true` flag marks which one matched your search.

5.4 Download Certificate PDF

Request

```
GET /api/v1/certificates/24-0366/download
X-API-Key: YOUR_API_KEY
```

Returns the PDF file with `Content-Type: application/pdf`.

6. Error Handling

All errors return a consistent JSON structure:

```
{
  "success": false,
  "error": "Description of what went wrong"
}
```

Code	Error	Resolution
`401`	Missing or invalid API key	Ensure `X-API-Key` header is present with a valid key
`403`	Insufficient permissions	Your key may not have access to this endpoint

Code	Error	Resolution
`404`	Certificate or product ID not found	Verify the certificate number or ASIN exists and is Active
`429`	Rate limit exceeded	Default: 200 API calls/hour (each call can return up to 100 certificates). Wait and retry with exponential backoff
`500`	Server error	Contact GCC support with the request details

7. Pagination

How It Works

The API paginates by **certificate**, not by product or ASIN. Each page returns up to **100 certificates per request**, and each certificate includes **all** of its associated ASINs and product identifiers in a single response — product IDs are never split across pages.

> **Example:** If a single certificate covers 50 ASINs, all 50 ASINs are returned with that certificate in one response. A page of 100 certificates could contain thousands of ASINs.

Capacity

Parameter	Value
Max certificates per page	100
Partner API rate limit	200 calls/hour
Max certificates per hour	20,000 (200 calls × 100 per page)
ASINs per certificate	Unlimited — all included in the response

For example, if Amazon has 300 active certificates covering 11,000+ ASINs, that requires only **3 API calls** (3 pages × 100 certificates) to retrieve the complete dataset — well within the 200 calls/hour limit. Rate limits reset at the top of each hour.

Pagination Example

```
// Page 1
GET /api/v1/certificates?pageNumber=1&pageSize=100

// Check response
"pagination": {
  "totalCount": 250,
  "pageNumber": 1,
  "pageSize": 100,
  "totalPages": 3,
  "hasNext": true,      // ← more pages available
  "hasPrevious": false
}

// Page 2
GET /api/v1/certificates?pageNumber=2&pageSize=100

// Continue until hasNext is false
```

Loop until `hasNext` is `false` to retrieve all certificates.

> **Page size & rate limits are configurable.** The default maximum of 100 certificates per page and 200 API calls per hour are suitable for most integration scenarios. If your volume requires higher throughput, contact GCC support to adjust these limits for your API key.

8. Security & Access Scope

Control	Details
Transport	HTTPS/TLS required for all requests
Authentication	API key via `X-API-Key` header
Data Isolation	Your key only returns certificates where Partner = "Amazon"; no other partner data is accessible
Status Filter	Only active certificates are returned; draft, expired, suspended, and revoked certs are excluded
Rate Limiting	200 API calls/hour (each call returns up to 100 certificates; configurable per key)
Key Management	Keys issued and revocable by GCC administrators

> **You will never see:** other partner data, internal notes, workflow history, draft certificates, or GCC user information.

9. Field Reference

Standard JSON vs GS1 JSON-LD Field Mapping

Data	Standard JSON Path	GS1 JSON-LD Path
Certificate Number	`data[].certificateNumber`	`products[].certificationInfo.certificateId`
Status	`data[].status` ("Active")	`certificationInfo.status` ("ACTIVE")
Issue Date	`data[].issueDate` (ISO 8601+time)	`issueDate` (date only)
Expiration Date	`data[].expirationDate`	`certificationInfo.endDate`
Cert Results	`data[].certificationResults`	`certificationInfo.value`
Cert Type	`data[].certificationType`	`certificationInfo.program`
Amazon Code	`data[].amazonCode`	`certificationInfo.amazonCode`
ASIN	`data[].productIds[].value`	`products[].additionalTradeItemId[].value`
GTIN	`data[].productIds[].value`	`products[].gtin` (14-digit padded)
PDF Link	`data[].pdfDownloadUrl`	*(use Standard endpoint)*

Product Identifier Types

Type	Format	Example
ASIN	10 characters (alphanumeric)	`B08XQRT456`
UPC	12 digits	`012345678901`
GTIN	14 digits	`00012345678901`
EAN	13 digits	`4006381333931`

Note: Only what is available will be passed; not all identifier types will be provided. At a minimum, ASIN will be included.

Available Sort Fields

sortBy Value	Description
`certificateNumber`	Sort by certificate number
`issueDate`	Sort by issue date
`expirationDate`	Sort by expiration date
`productName`	Sort by product name
`company`	Sort by company

GCC Support Contact

For API key requests, issues, or questions: Contact GreenCircle Certified.

Appendix A: GS1 JSON-LD Format Reference

Why Two Response Formats?

The GCC API provides two response formats for certificate data:

Format	Endpoint	Purpose
Standard JSON	`/certificates`	GCC's own field naming convention. Simple and direct.
GS1 JSON-LD	`/certificates/gs1`	Follows the GS1 Certification Data Model specification. Industry-standard format for interoperability.

Both formats return the same underlying data. The difference is the field names and structure.

What is GS1?

GS1 is the global standards organization that governs product identification across the supply chain — including UPC barcodes, GTINs, and EAN codes used by over 2 million companies worldwide. GS1 is a not-for-profit headquartered in Brussels with member organizations in over 110 countries.

GS1 US, the American member organization, published the **Certification Data Model** — a standardized format for exchanging product certification information between certification bodies and retailers. This is the specification the `/certificates/gs1` endpoint follows.

Why Do the GS1 Field Names Differ from Standard JSON?

The GS1 JSON-LD format uses field names defined by the **GS1 Web Vocabulary**, a formal specification maintained by GS1 Global. These field names cannot be changed because:

- The response references a published JSON Schema (`cert_schema.json`) and JSON-LD context (`cert_context.json`) hosted by GS1 US
- The schema enforces required field names, data types, and validation rules (e.g., `gtin` must be exactly 14 digits, `status` must be `ACTIVE` or `INACTIVE`)
- Any consumer validating against the schema would reject or ignore non-standard field names
- JSON-LD consumers map field names to GS1 vocabulary URIs — renamed fields would lose their semantic meaning

Complete Field Name Mapping

The following table shows how each piece of data maps between the two formats:

Data	Standard JSON Field	GS1 JSON-LD Field	GS1 Spec Definition
Product Name	`productName`	`name`	"Consumer friendly short description of the product"
Certificate Number	`certificateNumber`	`certificateId`	"A vendor specific identifier for the certificate"
Certification Type	`certificationType`	`program`	"The certification name or umbrella initiative"

Data	Standard JSON Field	GS1 JSON-LD Field	GS1 Spec Definition
Certification Results	`certificationResults`	`value`	"The certification's standard value or level"
Issue Date	`issueDate` (ISO 8601 + time)	`startDate` (date only)	"First date of validity for the certification"
Expiration Date	`expirationDate` (ISO 8601 + time)	`endDate` (date only)	"Last date of validity for the certification"
Countries	`certificationCountries`	`targetCountries`	"Countries where the certification applies"
Status	`status` ("Active")	`status` ("ACTIVE")	GS1 requires uppercase: `ACTIVE` or `INACTIVE`
Issuer	`issuer`	`agency`	"The certification agency or issuing body"
ASIN / Product IDs	`productIds[]`	`additionalTradeItemId[]`	GS1 standard for non-GTIN identifiers
GTIN	`productIds[].value`	`gtin`	14-digit padded Global Trade Item Number
PDF Link	`pdfDownloadUrl`	*N/A — use Standard endpoint*	Not part of GS1 spec

GCC Custom Extensions

The following fields are **GCC-specific extensions** included in the GS1 response for Amazon integration. They are not part of the GS1 specification but are safe to include — the GS1 schema allows additional properties.

Field	Description	Notes
`amazonCode`	Amazon's standardized certification identifier	Used by Amazon to map certificates to Climate Pledge Friendly badge categories
`pathway`	Certification pathway	Not applicable for all certification types; blank if not relevant

Governance & Schema References

Resource	URL
GS1 Web Vocabulary (global standard)	`gs1.org/voc/`
GS1 CertificationDetails class	`gs1.org/voc/CertificationDetails`
Certification Data Model — JSON-LD Context	`gs1us-technology.github.io/certification-data-model/cert_context.json`
Certification Data Model — JSON Schema	`gs1us-technology.github.io/certification-data-model/cert_schema.json`

> **Recommendation:** For most Amazon integration use cases, the GS1 endpoint (`/certificates/gs1`) is recommended. It provides an industry-standard format that ensures interoperability and future compatibility as GS1 certification exchange becomes more widely adopted across the retail supply chain.

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